

Evaluation of Binary Classifiers

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Evaluation of Binary Classifiers

Evaluation is important:

- models have to predict classes of new unlabeled data
- sometimes it's an integral part of the training process (e.g. in Decision Tree (Data Mining) for pruning) (see Cross Validation)
- also it's needed when we want to compare two or more different models (see Meta Learning)

Baseline

So for evaluating a classifier we need to set some baseline

- base rate
 - accuracy of a trivial classifier
 - the one that always predicts the majority class
- random rate
 - accuracy of random guess
 - need to have some domain knowledge to assign Random Distribution

Skewed Classes

- Suppose we have a binary classifier, e.g. cancer prediction.
 - We built some classification model $h_{\theta}(x)$
 - if we have $h_{\theta}(x) = 1$, we predict cancer, and if $h_{\theta}(x) = 0$, we predict no cancer.
- Then we find out that we have 1% errors for our classifier on test set, and 99% were correctly diagnosed
 - So the *error rate* is 1%
- But now suppose only 0.5% of patients have cancer
 - This is a *skewed class* - it's a tiny portion of another class

- We would predict better by always returning 0 (by using the trivial classifier)
 - (we'll have 0.5% error which is better than 1%)
- $\Rightarrow$ We need different evaluation metrics, not just error rate

Confusion Matrix

Confusion matrix is a 2×2 Contingency Table

- We divide our predictions and mis-predictions into this matrix

		Diagnostic Testing Measures [1]		
		Actual Class y		
		Positive	Negative	
$h_\theta(x)$ Test outcome	Test outcome positive	True positive (TP)	False positive (FP, Type I error)	Precision = $\frac{\#TP}{\#TP + \#FP}$
	Test outcome negative	False negative (FN, Type II error)	True negative (TN)	Negative predictive value = $\frac{\#TN}{\#FN + \#TN}$
		Sensitivity = $\frac{\#TP}{\#TP + \#FN}$	Specificity = $\frac{\#TN}{\#FP + \#TN}$	Accuracy = $\frac{\#TP + \#TN}{\#TOTAL}$

Main values of this matrix:

- **True Positive** - we predicted "+" and the true class is "+"
- **True Negative** - we predicted "-" and the true class is "-"
- **False Positive** - we predicted "+" and the true class is "-" (Type I error)
- **False Negative** - we predicted "-" and the true class is "+" (Type II error)
- (see also Statistical Tests of Significance#Type I and Type II Errors)

The following measures can be calculated:

- Accuracy
- Misclassification Error (or Error Rate)
- Positive predictive value (or precision)

$$P = \frac{TP}{TP + FP}$$

- Negative predictive value
- True Positive Rate (also Sensitivity or Recall)
 - Fraction of positive examples correctly classified

$$tpr = \frac{TP}{TP + FN}$$

- False Positive Rate (also Fall-Out)

- Fraction of negative examples incorrectly classified

- $$\text{fpr} = \frac{\text{FP}}{\text{FP} + \text{TN}}$$

- Specificity
- Support - fraction of positively classified examples

- $$\text{sup} = \frac{\text{TP} + \text{FP}}{N} = \frac{\text{predicted pos}}{\text{total}}$$

Accuracy and Error Rate

In practice, these are the most widely used metrics

- Accuracy:
$$\text{acc} = \frac{\text{TP} + \text{TN}}{N}$$
 - fraction of correctly classified examples
- Error Rate:
$$\text{error} = \frac{\text{FN} + \text{FP}}{N} = 1 - \text{acc}$$
 - Fraction of misclassified examples

Precision

For all input data that we predicted $h_{\theta}(x) = 1$ what fraction actually have $y = 1$?

$$P = \text{Precision} = \frac{\# \text{ TP}}{\# \text{ predicted positives}} = \frac{\# \text{ TP}}{\# \text{ TP} + \# \text{ FP}}$$

- Out of all the people we thought have cancer, how many actually had it?
- High precision is good
- we don't tell many people that they have cancer when they actually don't

Recall

For all input data that actually have $y = 1$, what fraction did we correctly detect as $h_{\theta}(x) = 1$?

$$R = \text{Recall} = \frac{\# \text{ TP}}{\# \text{ actual positives}} = \frac{\# \text{ TP}}{\# \text{ TP} + \# \text{ FN}}$$

- Out of all the people that do actually have cancer, how much we identified?
- The higher the better:
- We don't fail to spot many people that actually have cancer

- For a classifier that always returns zero (i.e. $h_{\theta}(x) = 0$) the Recall would be zero
- That gives us more useful evaluation metric

- And we're much more sure

The F Measure is a combination of Precision and Recall

Example

Diagnostic Testing Wikipedia Example [2]				
		Patients with bowel cancer (as confirmed on endoscopy)		
		Positive	Negative	
Fecal Occult Blood Screen Test Outcome	Test Outcome Positive	True Positive (TP) = 20	False Positive (FP) = 180	Positive predictive value = $TP / (TP + FP)$ = $20 / (20 + 180)$ = 10%
	Test Outcome Negative	False Negative (FN) = 10	True Negative (TN) = 1820	Negative predictive value = $TN / (FN + TN)$ = $1820 / (10 + 1820)$ ≈ 99.5%
		Sensitivity = $TP / (TP + FN)$ = $20 / (20 + 10)$ ≈ 67%	Specificity = $TN / (FP + TN)$ = $1820 / (180 + 1820)$ = 91%	

Visual Analysis

Visual ways of evaluating the performance of a classifier

- ROC Analysis - True Positive Rate vs False Positive Rate
- Cumulative Gain Charts - True Positive Rate vs Predicted Positive Rate

Not Binary Classifiers

When we have multi-class classifiers we can use:

- Contingency Table
 - just show misclassified examples side-by-side
- Cost Matrix
 - we define the cost for each misclassification
 - and calculate the total cost
- some measures can be extended to multiclass classifiers:
 - see Evaluation of Multiclass Classifiers